

B. JOEL GONZALEZ

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EDUCATION

Carnegie Mellon University (CMU)

Pittsburgh, PA

College of Engineering

2021 - Present

- Ph.D. Candidate in *Electrical and Computer Engineering*
- Advisor: Dr. L. Richard Carley
- Interests: integrated systems design, nanofabrication, metrology and instrumentation
- M.S. in *Electrical and Computer Engineering*, concentration in *Integrated Systems* — May 2024

University of Pennsylvania (UPenn)

Philadelphia, PA

School of Engineering & Applied Science

2015 - 2021

- M.S.E. in *Robotics* — May 2021
- B.S.E. in *Computer Science*, minor in *Mathematics* — December 2019

TEACHING ASSISTANTSHIPS

Asterisks denote semesters as lead TA.

CMU:

- 18-723: *RFIC Design and Implementation* — Spring 2026, Fall 2026
- 18-427/727: *Board-Level RF Systems for IoT* — Fall 2022*, Fall 2023*, Fall 2024*, Fall 2025*

UPenn:

- ESE350/ESE519: *Embedded Systems Lab* — Fall 2020, Spring 2021*
- CIS371/CIS501: *Computer Architecture* — Fall 2019, Spring 2020
- CIS380/CIS548/CIT595: *Operating Systems* — Fall 2019, Spring 2020*, Summer 2020*
- CIS240/CIT593: *Introduction to Computer Systems* — Fall 2018, Spring 2019, Summer 2019

WORKS

Asterisks denote works where authors contributed equally.

Publications and Preprints

- B.J. Gonzalez, T.J. Zajdel, L.R. Carley. "Leveraging Inexpensive Lab Tools for Hands-on RF Systems Course". arXiv preprint (2026). <https://doi.org/10.48550/arXiv.2607.23911>
- B.J. Gonzalez*, Y.I. Chen*, J.K. Wirant, J.S. Kunselman, E. Bourcart, M.T. Moneck, T. Srimani, and L. Pileggi, "The Hacker Fab: An Open-Source Initiative for Nanofabrication Education," 2026 IEEE International Symposium on Circuits and Systems (ISCAS), Shanghai, China, 2026, pp. 3758-3762, doi: 10.1109/ISCAS66217.2026.11562797 [Lecture].
- S. Hussain, B.J. Gonzalez, L. Chomas, and L.R. Carley. "Anisotropic high-aspect-ratio etching of sub-100-nm-diameter columns in silicon oxide." MRS Advances (2026). <https://doi.org/10.1557/s43580-026-01684-7>
- B.J. Gonzalez, E. Bourcart, J.K. Wirant, M. Juan, J. Wang, M.T. Moneck. "A low-cost, open-source maskless photolithography stepper for microfabrication". arXiv preprint (2025). <https://doi.org/10.48550/arXiv.2510.15082>

Presentations

- B.J. Gonzalez*, Y.I. Chen*, J.K. Wirant, J.S. Kunselman, E. Bourcart, M.T. Moneck, T. Srimani, and L. Pileggi, "Live Demonstration: The Hacker Fab Maskless Photolithography Stepper," 2026 IEEE International Symposium on Circuits and Systems (ISCAS), Shanghai, China, 2026, pp. 1684-1684, doi: 10.1109/ISCAS66217.2026.11563033 [Poster].
- Z. Liu et al., "Ferromagnetic Resonance Measurement of Nanoscale Thin Films Using a Lab-on-Chip Interferometer," 2026 IEEE International Magnetic Conference - Short Papers (INTERMAG Short Papers), Manchester, United Kingdom, 2026, pp. 1-2, doi: 10.1109/INTERMAGShortPapers68882.2026.11596350 [Poster].
- S. Hussain, B.J. Gonzalez, L. Chomas, and L.R. Carley (2025, December 2). "Anisotropic High Aspect Ratio Etching of Sub-100nm Diameter Columns in Silicon Oxide". Materials Research Society Fall 2025 Meeting, Boston, MA, United States [Poster].
- B.J. Gonzalez, M.T. Moneck, and T. Srimani (2025, March 21-23). "The Hacker Fab: A DIY Approach to Nanofabrication". 2025 ECEDHA Annual Conference & ECEExpo, Norfolk, VA, United States [Exhibition].

EXPERIENCE

MIT Lincoln Laboratory

Lexington, MA

Division of Communication Systems

May 2023 - August 2023

- Summer 2023 research intern working on RF systems design and analysis of communication systems
- Used Python programming and RF hardware to analyze communication networks, spectra, and phase noise
- Internship completed as part of employer sponsorship via GEM Fellowship

ACTIVITIES, AWARDS, AND RECOGNITIONS

CMU:

- Carnegie Institute of Technology Dean's Fellow
- Recipient of the *GEM Fellowship*, with sponsorship from *MIT Lincoln Lab* for Summer 2023
- Completed the *Future Faculty Program* in Fall 2022 (Eberly Center for Teaching Excellence and Educational Innovation)
- Selected to participate in the 2022 *NextProf Pathfinder* workshop hosted by UMich and UCSD
- 2022-25 *CMU Robotics Club* Officer

UPenn:

- 2019-20 *CIS J.P. Eckert Fellow*
- 2020 inductee of the *CIS Max Mintz Undergraduate TA Hall of Fame*
- Honorable Mention for the 2021 Penn Engineering Outstanding Teaching Award
- Recipient of the *Summer 2020 TA Award for Excellence in Student Support with Distinction*
- Recipient of the 2019 *Littlejohn Scholars Summer Research Grant*
- Recipient of the 2018 *Penn Engineering Exceptional Service Award*
- Recipient of the 2017 *Penn Undergraduate Research Mentoring (PURM) Grant*